

Product Recalls and Channel Pricing



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Abstract We propose a stochastic differential game between a manufacturer and a retailer to investigate how the risk of facing a product recall impacts pricing strategies in marketing channels. By doing so, we analyze whether vendor agreements between manufacturers and retailers, which are signed before any unit is sold, could distort channel profits by aggravating double marginalization. We characterize the equilibrium pricing strategies in closed form for both linear and quadratic costs of recall. We find that the manufacturer and the retailer respond differently to certain clauses of the vendor agreement, but that in equilibrium, such agreements do not distort channel profit, even when costs of recall are quadratic.

Keywords Marketing channel · Pricing · Product recall · Stochastic differential games

1 Introduction

Product recalls hurt not only manufacturers, but also the retailers that carry the recalled products. To protect themselves against recall costs, retailers often require sellers to sign vendor agreements that define ex ante how manufacturers will reimburse retailers' expenses related to recalls. In the USA for instance, Safeway specifies that "Seller shall be responsible for all reasonable costs and expenses associated with the recall."¹ Other retailers are even more specific and provide a breakdown of recalls costs, which are usually divided into two categories, i.e., cost of goods sold and other administrative and logistical costs. Albertsons companies,

¹<https://suppliers.safeway.com>.

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for example, define the cost of goods as “number of units $\times$ cost,” whereas Kroger details that “Cost of goods will be billed using the most recent FCB cost times the number of retail units submitted through the Quick Recall application.”² Meanwhile, Albertsons companies specify that recall/withdrawal logistics costs include retailer labor fee (\$40/*store*), reclamation fee (\$0.84/*unit*), hazardous material fee (\$0.84/*unit*), and warehouse recovery fee (\$40/*case*).

Even though such contractual agreements are common in practice, there is no study, to the best of our knowledge, that investigates how they would impact channel pricing strategies and most importantly profits. Despite being rich, the literature on product recalls (e.g., Dawar and Pillutla 2000; Roehm and Tybout 2006; Van Heerde et al. 2007; Rubel et al. 2011; Liu and Shankar 2015; Borah and Tellis 2016; Eilert et al. 2017; Rubel 2017) does not provide any insight on how manufacturers and retailers should strategically set prices in marketing channels when product recalls can occur. In particular, it is unclear how contractual agreements demanded by retailers from manufacturers improve or degrade profits.

To investigate such issues, we propose a stochastic differential game between a manufacturer and a retailer. The game is stochastic because the time at which the recall will take place is unknown. In such contexts, the manufacturer sets the wholesale price and the retailer sets the retail price paid by consumers. The proposed model considers two state variables. The first state variable is the number of products that must be recalled when the crisis occurs, as in Rubel (2017), which will determine the logistics costs faced by the manufacturer and the retailer. The second state variable keeps tracks of the financial transfer that will take place between the channel partners with respect to the “cost of goods” clause.

We derive the optimal pricing strategies and value functions in closed forms for both linear and quadratic costs, which allows to obtain several insights. We first find that the manufacturer and the retailer respond differently to clauses in the vendor agreement. For instance, we find that the equilibrium retail price does not vary whether the “cost of goods” clause is included or not in the vendor agreement, whereas the wholesale price does. Furthermore, we find that when the costs of recalls are linear, the optimal pricing strategies are constant over time and do not depend on accumulation of units that would have to be recalled when a crisis happens. Conversely, when the costs of recalls are quadratic, firms should implement dynamic pricing strategies that vary with the number of units that would have to be recalled when a crisis happens. We find, however, that in this case the retailer and the manufacturer could respond differently. Specifically, while the retailer would always increase the retail price as the number of units to be recalled increases, we find that the manufacturer can increase or decrease the wholesale price. Finally, we find that vendor agreements do not alter total channel profits, but serve as a way to allocate costs between the two channel partners. Our results thus add to the literature on product recalls by considering the decentralized channel structure and its impact on pricing strategies when envisioning a product harm crisis.

²<https://www.thekrogerco.com/>.

We also contribute to the differential games literature in marketing (e.g., Jørgensen and Zaccour 2004; Breton et al. 2006; Martin-Herran et al. 2008; Rubel 2013) and in particular on marketing channels (e.g., Jørgensen et al. 2000; Rubel and Zaccour 2007; Zaccour 2008) by investigating the interplay between pricing strategies and uncertain product recalls.

In the next section, we present the model. In Sect. 3, we present our results when costs of recall are linear and in Sect. 4, we provide new insights when costs of recall are quadratic. Finally, we conclude in Sect. 5.

2 Model

We consider a dynamic marketing channel comprised of a manufacturer and a retailer. Time (t) is continuous with $t \in [0, T]$, where T is the uncertain time at which the manufacturer's products have to be recalled because of the potential harm products could impose on consumers. Thus T is a random variable that is characterized by the stochastic process $\{\Gamma(t) : t \geq 0\}$ defined as follows:

$$\lim_{dt \rightarrow 0} \frac{P[\Gamma(t + dt) = 1 | \Gamma(t) = 0]}{dt} = \chi, \quad \lim_{dt \rightarrow 0} \frac{P[\Gamma(t + dt) = 0 | \Gamma(t) = 1]}{dt} = 0, \quad (1)$$

where χ is the intensity of the jump process governing T (see, e.g., Haurie and Moresino 2006).

Demand for the manufacturer's product $q(p)$ is assumed to be linear such that

$$q(p) = v - \alpha \times p, \quad (2)$$

with $v > 0$, $\alpha > 0$ and where p is the retail price chosen by the retailer. Our first state variable is the volume of units that must be recalled at the time of recall, i.e. $x(T)$, such that its dynamics is

$$\frac{dx}{dt} = q(p) - \delta x(t), \quad (3)$$

with $x(0) = 0$. Similar to Rubel (2017), we note that the total number of units to be recalled is not necessarily equal to cumulative sales due to possible attrition, which we capture through $\delta > 0$. The second state variable that we consider is the financial transfer that the manufacturer will pay to the retailer to cover the costs of goods sold, i.e., $B(T)$, such that

$$\frac{dB}{dt} = q(p) \times w \times \mathbb{E}(\theta(t)), \quad (4)$$

with $B(0) = 0$ and where w is the manufacturer's wholesale price. Furthermore, $\mathbb{E}(\theta(t))$ captures the expected proportion of goods sold at time t , i.e., $q(\cdot)$, that will

be recalled. Specifically, due to the parameter δ in (3), not all products sold will be recalled; the volume of goods recalled depends on δ and the *uncertain* time at which the recall is issued such that $\theta(t) = q(p(t)) \times e^{-\delta(T-t)}$, where T and consequently $T - t$ is a random variable. As a result, $\mathbb{E}(\theta(t)) = \frac{\chi}{\delta + \chi}$, which is equal to 1 when $\delta = 0$, as it should be.

The manufacturer's and retailer's objective functions are defined as

$$\Pi_M(t, x, B) = \mathbb{E}_\Gamma \left\{ \int_0^T e^{-rt} q(p) \times (w - c) dt - (B(T) + \kappa_M x(T)) e^{-rT} \right\}, \quad (5)$$

and

$$\Pi_R(t, x, B) = \mathbb{E}_\Gamma \left\{ \int_0^T e^{-rt} q(p) \times (p - w) dt - (-B(T) + \kappa_R x(T)) e^{-rT} \right\}, \quad (6)$$

respectively, where r is the discount rate, c is the manufacturer's marginal cost of production, and κ_i is the cost parameter of player $i = \{M, R\}$ related to the recall. Recall that an important clause of vendor agreements pertains to the logistical costs engendered by the recall of x units. Specifically, such clauses specify how much of the retailer's recall costs will be reimbursed by the manufacturer. Formally, such a clause is equivalent to writing $\kappa_M = k_M + \phi k_R$ and $\kappa_R = (1 - \phi)k_R$, where k_M and k_R are the actual costs incurred by both channel partners and where $0 < \phi < 1$ is the share of the retailer's costs that is reimbursed by the manufacturer. For instance, setting ϕ to one, i.e., the manufacturer covers all the costs of the retailers, implies that $\kappa_R = 0$.

To solve the stochastic problems faced by the manufacturer and the retailer, we rewrite their respective objective functions as infinite horizon deterministic problems via integration by parts (see, e.g., Haurie and Moresino 2006; Rubel 2013), such that (5) and (6) become

$$\Pi_M(x, B) = \int_0^\infty e^{-(r+\chi)t} \{q(p) \times (w - c) - \chi (B + \kappa_M x)\} dt, \quad (7)$$

and

$$\Pi_R(x, B) = \int_0^\infty e^{-(r+\chi)t} \{q(p) \times (p - w) - \chi (-B + \kappa_R x)\} dt, \quad (8)$$

respectively. Assuming that the manufacturer and the retailer play Stackelberg, with the retailer as the follower, we define the retailer's value function as $J_R(x) = \underset{p}{\text{Max}} \Pi_R$ for any wholesale price w announced by the manufacturer. Consequently, the retailer's Hamilton-Jacobi-Bellman (HJB) equation is

$$(r + \chi)J_R(x, B) = \underset{p}{Max} \left\{ q(p) \times (p - w) - \chi(-B + \kappa_R x) \right. \\ \left. + \frac{\partial J_R}{\partial B} \left(q(p)w \frac{\chi}{\delta + \chi} \right) + \frac{\partial J_R}{\partial x} (q(p) - \delta x) \right\}. \quad (9)$$

Differentiating (9) with respect to p and equating the resulting equation to zero yields that the retailer's optimal pricing strategy is

$$p^* = \frac{v + w \times \alpha}{2\alpha} - \frac{1}{2} \times \left(w \times \frac{\chi}{\delta + \chi} \times \frac{\partial J_R}{\partial B} + \frac{\partial J_R}{\partial x} \right), \quad (10)$$

where $\frac{v + w \times \alpha}{2\alpha}$ is the static retail price and $\frac{1}{2} \times \left(w \times \frac{\chi}{\delta + \chi} \times \frac{\partial J_R}{\partial B} + \frac{\partial J_R}{\partial x} \right)$ is the additional term that adjusts the retail price in anticipation of the product recall. Next, we replace the retailer's pricing strategy (10) in the manufacturer's value function $J_M(x) = \underset{w}{Max} \Pi_M$, such that the manufacturer's HJB equation is

$$(r + \chi)J_M(x, B) = \underset{w}{Max} \left\{ q(p^*) \times (w - c) - \chi(B + \kappa_M x) \right. \\ \left. + \frac{\partial J_M}{\partial B} \left((v - \alpha p^*) \frac{w\chi}{\delta + \chi} \right) + \frac{\partial J_M}{\partial x} (v - \alpha p^*) \right\}. \quad (11)$$

Differentiating (11) with respect to w and equating the resulting expression to zero yields that

$$w^* = \frac{v + \alpha \left(c + \frac{\partial J_R}{\partial x} - \frac{\partial J_M}{\partial x} \right) + \frac{\chi}{\delta + \chi} \times \left(\frac{\partial J_M}{\partial B} \left(v + \alpha \frac{\partial J_R}{\partial x} \right) + \alpha \frac{\partial J_R}{\partial B} \left(\frac{\partial J_M}{\partial x} - c \right) \right)}{2\alpha \times \left(\frac{\chi}{\delta + \chi} \times \frac{\partial J_R}{\partial B} - 1 \right) \times \left(\frac{\chi}{\delta + \chi} \frac{\partial J_M}{\partial B} + 1 \right)}. \quad (12)$$

3 Results

To characterize the retailer's value function, we first replace (10) in (8), then replace w by w^* in the resulting expression. Similarly, to characterize the manufacturer's value function, we first replace (10) in (11), then replace w by w^* in the resulting expression. Next, owing to its structure, we note that the game played by the retailer and the manufacturer is of the linear-state variety and thus conjecture that

$$J_M(x, B) = M_1 + M_2 B + M_3 x \quad \text{and} \quad J_R(x, B) = R_1 + R_2 B + R_3 x, \quad (13)$$

where $M_1, M_2, M_3, R_1, R_2,$ and R_3 solve the following system of equations

$$\begin{aligned}
 (r + \chi)M_1 &= \\
 &- \frac{(\alpha(c(\frac{\chi}{\delta+\chi}R_2 - 1) + \frac{\chi}{\delta+\chi}M_2R_3 + \frac{\chi}{\delta+\chi}M_3(-R_2) + M_3 + R_3) + \frac{\chi}{\delta+\chi}M_2v + v)^2}{8\alpha(\frac{\chi}{\delta+\chi}M_2 + 1)(\frac{\chi}{\delta+\chi}R_2 - 1)} \\
 (r + \chi)M_2 &= -\chi \\
 (r + \chi)M_3 &= -M_3\delta - \kappa_M\chi \\
 (r + \chi)R_1 &= \\
 &\frac{(\alpha(c(\frac{\chi}{\delta+\chi}R_2 - 1) + \frac{\chi}{\delta+\chi}M_2R_3 + \frac{\chi}{\delta+\chi}M_3(-R_2) + M_3 + R_3) + \frac{\chi}{\delta+\chi}M_2v + v)^2}{16\alpha(\frac{\chi}{\delta+\chi}M_2 + 1)^2} \\
 (r + \chi)R_2 &= \chi \\
 (r + \chi)R_3 &= -R_3\delta - \kappa_R\chi.
 \end{aligned}$$

As a result, we obtain the following proposition.

Proposition 1 *The manufacturer's and retailer's value functions are*

$$J_M(x, B) = M_1 + M_2B + M_3x \quad \text{and} \quad J_R(x, B) = R_1 + R_2B + R_3x, \quad (14)$$

$$\text{with } M_1 = \frac{(\alpha c(\delta + \rho + \chi) + \alpha \chi(\kappa_M + \kappa_R) - v(\delta + \rho + \chi))^2}{8\alpha(\rho + \chi)(\delta + \rho + \chi)^2}, M_2 = -\frac{\chi}{\delta + \chi}, M_3 = -\frac{\kappa_M\chi}{\delta + \chi + \rho}, R_1 = \frac{M_1}{2}, R_2 = \frac{\chi}{\delta + \chi} \text{ and } R_3 = -\frac{\kappa_R\chi}{\delta + \chi + \rho}.$$

As a result, the optimal wholesale and retail prices are

$$w^* = \frac{(\delta + \chi)(\rho + \chi)((v + \alpha c)(\delta + \rho + \chi) + \alpha \chi(\kappa_M - \kappa_R))}{2\alpha(\delta + \rho + \chi)(\delta(\rho + \chi) + \rho\chi)} \quad (15)$$

and

$$p^* = \frac{\alpha c(\delta + \rho + \chi) + \alpha \chi(\kappa_M + \kappa_R) + 3v(\delta + \rho + \chi)}{4\alpha(\delta + \rho + \chi)}, \quad (16)$$

respectively.

Corollary 1 *If the vendor agreement were not to include a clause on the cost of goods sold, the manufacturer's and retailer's value functions would be such that $M_2 = R_2 = 0$, and as a result, the optimal pricing strategies become*

$$\tilde{w}^* = \frac{(v + \alpha c)(\delta + \rho + \chi) + \alpha \chi (\kappa_M - \kappa_R)}{2\alpha(\delta + \rho + \chi)} \quad (17)$$

and

$$\tilde{p}^* = \frac{\alpha c(\delta + \rho + \chi) + \alpha \chi (\kappa_M + \kappa_R) + 3v(\delta + \rho + \chi)}{4\alpha(\delta + \rho + \chi)}. \quad (18)$$

Proposition 1 and its corollary allow us to derive several insights. We first learn that the manufacturer and the retailer respond differently to the clauses of the vendor agreement since the optimal pricing strategies are such that the *optimal* retail price should not change due to the inclusion of the cost of goods clause, whereas the optimal wholesale price should; formally,

$$\tilde{p}^* = p^* \quad \text{whereas} \quad \tilde{w}^* < w^*.$$

From a managerial perspective, these two relationships mean that even though the manufacturer increases the wholesale price in response to the inclusion of the clause on the cost the goods sold, the retailer does not. This result is surprising as economic intuition would have suggested that the retailer should have instead increased the retail price in response to a higher wholesale price.

The second new insight that Proposition 1 reveals is that the vendor agreement does not distort the total channel profit $J_D = J_M + J_R$. Instead, it only reallocates profit (and costs) between the two channel partners in case of a recall, as it should be. To see this point, we first consider the impact of clause related to the cost of goods sold on the total channel profit. Specifically,

$$J_D = \frac{3}{2}M_1 - (k_M + k_R)\frac{\chi}{\delta + r + \chi}x,$$

which implies that $\frac{\partial J_D}{\partial B} = 0$. Furthermore, we recall that the second clause that is usually included in vendor agreements pertains to the logistical costs engendered by the recall of x units and how much of the retailer's costs are supported by the manufacturer, i.e., $0 < \phi < 1$. In equilibrium, it is interesting to note that

$$\frac{\partial J_D}{\partial \phi} = \frac{dJ_D}{d\phi} = 0,$$

not only because $\frac{\partial M_1}{\partial \phi} = 0$, but also because $\frac{\partial x}{\partial \phi} = 0$ since the equilibrium price p^* does not vary with ϕ . Therefore, one important finding from Proposition 1 and its corollary is that vendor agreements *do not distort* channel profit by aggravating double marginalization, a finding that is absent from the extant literature on channel.

4 Equilibrium Under Quadratic Recall Costs

We now analyze the case where costs of recall are quadratic in x , i.e., $\kappa_M x^2$ and $\kappa_R x^2$. As a result, the retailer's HJB equation, for any w , is

$$\begin{aligned} & (r + \chi)J_R(x, B) \\ &= \underset{p}{Max} \left\{ q(p) \times (p - w) - \chi \left(-B + \kappa_R x^2 \right) + \frac{\partial J_R}{\partial B} \frac{dB}{dt} + \frac{\partial J_R}{\partial x} \frac{dx}{dt} \right\}. \end{aligned} \quad (19)$$

The first order condition for (19) yields that

$$p^* = \frac{v + w \times \alpha}{2\alpha} - \frac{1}{2} \times \left(w \times \frac{\chi}{\delta + \chi} \times \frac{\partial J_R}{\partial B} + \frac{\partial J_R}{\partial x} \right). \quad (20)$$

We then write the manufacturer's HJB as

$$\begin{aligned} (r + \chi)J_M(x, B) &= \underset{w}{Max} \left\{ q(p^*) \times (w - c) - \chi \left(B + \kappa_M x^2 \right) \right. \\ &\quad \left. + \frac{\partial J_M}{\partial B} \left((v - \alpha p^*) \frac{w\chi}{\delta + \chi} \right) + \frac{\partial J_M}{\partial x} (v - \alpha p^*) \right\}. \end{aligned} \quad (21)$$

The first order condition for the manufacturer's optimal wholesale price yields that w^* is identical to (12), as it should be. We then replace the optimal strategies in the HJB equations and note that owing to the linear-quadratic structure of the Stackelberg game, we posit that

$$J_M(x, B) = N_0 + N_1 x + \frac{N_2}{2} x^2 + N_3 B \quad \text{and} \quad J_R(x, B) = S_0 + S_1 x + \frac{S_2}{2} x^2 + S_3 B, \quad (22)$$

where $(\rho + \chi)S_3 = \chi \implies S_3 = \frac{\chi}{\chi + \rho}$ and $(\rho + \chi)N_3 = -\chi \implies N_3 = -\frac{\chi}{\chi + \rho}$, and where the other coefficients solve the system of equations

$$\frac{\chi + \rho}{2} N_2 = \frac{\alpha (N_2 + S_2)^2}{8} - \delta N_2 - \kappa_M \chi \quad (23)$$

$$\frac{\chi + \rho}{2} S_2 = \frac{\alpha (N_2 + S_2)^2}{16} - \delta S_2 - \kappa_R \chi \quad (24)$$

$$(\chi + \rho) N_1 = \frac{(N_2 + S_2) (v + \alpha (N_1 + S_1 - c))}{4} - N_1 \delta \quad (25)$$

$$(\chi + \rho) S_1 = \frac{(N_2 + S_2) (v + \alpha (N_1 + S_1 - c))}{8} - S_1 \delta \quad (26)$$

$$(\chi + \rho)N_0 = \frac{(v + \alpha(N_1 + S_1 - c))^2}{8\alpha} \quad (27)$$

$$(\chi + \rho)S_0 = \frac{(v + \alpha(N_1 + S_1 - c))^2}{16\alpha}. \quad (28)$$

Proposition 2 *Under quadratic costs, the value functions of the manufacturer and the retailers are*

$$J_M(x, B) = N_0 + N_1x + \frac{N_2}{2}x^2 + N_3B \quad \text{and} \quad J_R(x, B) = S_0 + S_1x + \frac{S_2}{2}x^2 + S_3B, \quad (29)$$

respectively, with

$$N_3 = -\frac{\chi}{r + \chi}, \quad S_3 = -N_3, \quad N_2 = \frac{\alpha Y^2 - 8\kappa_M \chi}{4(2\delta + r + \chi)}, \quad S_2 = \frac{\alpha Y^2 - 16\kappa_R \chi}{8(2\delta + r + \chi)}, \quad (30)$$

$$N_1 = -\frac{2(v - \alpha c)(3\alpha Y^2 - 16\chi(\kappa_M + \kappa_R))}{9\alpha^2 Y^2 - 16(4(\delta + r + \chi)(2\delta + r + \chi) + 3\alpha\chi(\kappa_1 + \kappa_2))} \quad (31)$$

$$S_1 = \frac{N_1}{2}, \quad N_0 = \frac{(v - c\alpha + \alpha \frac{3N_1}{2})^2}{8\alpha(r + \chi)}, \quad S_0 = \frac{N_0}{2}, \quad (32)$$

$$\text{with } Y = \frac{4(2\delta + r + \chi - \sqrt{3(\kappa_M + \kappa_R)\alpha\chi + (2\delta + r + \chi)^2})}{3\alpha}. \quad (33)$$

As a result, the optimal pricing strategies are

$$w^* = \frac{(2(v + c\alpha) - N_1\alpha)(r + \chi)}{4\alpha r} + \frac{(S_2 - N_2)(r + \chi)}{2r} \times x \quad (34)$$

$$p^* = \frac{6v + 2c\alpha - 3N_1\alpha}{8\alpha} - \frac{N_2 + S_2}{4} \times x \quad (35)$$

Proof Applying the method of undetermined coefficients to identify the values of the coefficients revealed that Y could take two values, i.e.,

$$\frac{4(2\delta + r + \chi + \sqrt{3(\kappa_M + \kappa_R)\alpha\chi + (2\delta + r + \chi)^2})}{3\alpha}$$

and

$$\frac{4(2\delta + r + \chi - \sqrt{3(\kappa_M + \kappa_R)\alpha\chi + (2\delta + r + \chi)^2})}{3\alpha},$$

We keep the second root to ensure convergence to a stable steady state. ■

Proposition 2 displays how the manufacturer and the retailer should dynamically vary their prices based on the volume of units that should be recalled in the event of a crisis. Such pricing strategies depart from those obtained in Proposition 1 where optimal pricing strategies were constant. Given the feedback structure of the pricing strategies in Proposition 2, we learn that the manufacturer and the retailer could respond differently to changes in x . Specifically, it is apparent from Proposition 2 that the optimal retail price always increases as x increases since $N_2 + S_2 < 0$, i.e., $\frac{\partial p^*}{\partial x} > 0$. Conversely, however, the optimal wholesale price can either decrease or increase as x increases, since

$$S_2 - N_2 = \frac{4(3\alpha\chi(\kappa_M - 2\kappa_R) - (2\delta + r + \chi)^2)}{9\alpha(2\delta + r + \chi)} + \frac{4\sqrt{(2\delta + r + \chi)^2 + 3\alpha\chi(\kappa_M + \kappa_R)}}{9\alpha},$$

which could be positive or negative depending on parameters' values.³

Finally, Proposition 2 allows us to confirm that even when recall costs are quadratic in the number of units, the vendor agreement does not distort the total channel profit $J_D = J_M + J_R$, since $\frac{\partial J_D}{\partial B} = \frac{\partial J_D}{\partial \phi} = 0$.

5 Concluding Remarks

The proposed model informs how a manufacturer and a retailer should strategically price a product that is likely to be recalled at a future, yet uncertain, date. The proposed model allows us to analyze the role of vendor agreements on pricing strategies and most importantly profits. We find that the manufacturer and the retailer respond differently to the clauses found in such agreements. For instance, when recall costs are linear in the number of units that have to be recalled, we find that the retailer does not vary the retail price depending on whether the clause on cost of goods sold is included or not in the agreement, while the manufacturer does. When recall costs are quadratic, we find that the retailer always increases the retail price, while the manufacturer can either decrease or increase the wholesale price depending on parameters' values. Meanwhile, we are also able to conclude, based on the proposed model, that vendor agreements do not distort channel profits whether costs are linear or quadratic. This result adds to the marketing channel literature (see, e.g., Ingene et al. 2012) by considering how the risk of a major recall might affect optimal pricing strategies by a manufacturer and a retailer.

³For instance, when $\chi = 0.001$, $\rho = 0.05$, $\alpha = 1$, $\delta = 0.01$, $\kappa_M = 0.1$ and $\kappa_R = 0.1$, then $S_2 - N_2 < 0$; however, changing κ_M from 0.1 to 0.15 yields $S_2 - N_2 > 0$.

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